



XYLOFY AI

EMPLOYEE ATTRITION PREDICTION USING MACHINE LEARNING

PROJECT SUMMARY REPORT



BUILDING DATA-DRIVEN SOLUTIONS
TO RETAIN TALENT AND
STRENGTHEN ORGANIZATIONS.

PROJECT OVERVIEW

Project Title : Employee Attrition
Prediction Using Machine Learning

Domain : HR Analytics

Organization : Xylofy AI

Technology : Python, Pandas, Scikit-learn,
Matplotlib, Seaborn

Tools : Jupyter Notebook

Problem Type : Binary Classification

DEVELOPER DETAILS

Name : Pokala Ravindranath Tagoor

Intern ID : XYAI2026-0619

Program : Data Analytics &
Machine Learning Internship

Phone : 9493929098

Email : pokalaravindranathtagoor@gmail.com

Address : Hyderabad, Telangana, India

DATASET DETAILS

Dataset : IBM HR Analytics
Employee Attrition Dataset

Records : 1470 Employees

Features : 35 (Numerical + Categorical)

Target Variable : Attrition (Yes / No)

Target Distribution

No (Stayed)	1233 (83.88%)
Yes (Left)	237 (16.12%)

Attrition Rate : 16.12%

SUBMISSION DETAILS

Submission Date : 24/06/2026

Last Date of Submission : 30/06/2026

PROJECT AT A GLANCE – KEY METRICS



1470
Employees
Dataset Size



35
Features
(Numerical + Categorical)



3
Models Built
& Evaluated



16.12%
Attrition Rate
(Yes)
of Total Employees



0.93
Best ROC-AUC
(Gradient Boosting)



Best Model
Gradient Boosting
Classifier

1. PROJECT OBJECTIVE

To build a machine learning model that predicts whether an employee is likely to leave the company based on various factors like job role, salary, work-life balance, overtime, experience, and satisfaction levels. The goal is to help HR teams identify at-risk employees and take proactive retention actions.

2. DATA PREPROCESSING

- Handled missing values using appropriate imputation
- Encoded categorical variables using One-Hot Encoding
- Scaled numerical features using StandardScaler
- Removed irrelevant columns
- Handled class imbalance using class weight
- Split dataset into Train (80%) and Test (20%)

3. EXPLORATORY DATA ANALYSIS

- Analyzed attrition by department, job role, gender, age, and more
- Identified key patterns and trends in employee behavior
- Visualizations used to understand relationships between features and attrition



4. KEY FINDINGS

- Sales department has the highest attrition rate.
- Employees with lower income tend to leave more.
- Overtime, low job satisfaction, and poor work-life balance increase attrition risk.
- Most employees leave within the first 3 years.
- Laboratory Technician and Sales Executive roles have higher attrition.

5. MODEL BUILDING & EVALUATION

We built and evaluated 3 machine learning models.

Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC
Logistic Regression	0.85	0.78	0.68	0.73	0.88
Random Forest Classifier	0.90	0.86	0.77	0.81	0.92
Gradient Boosting Classifier	0.92	0.88	0.82	0.85	0.93



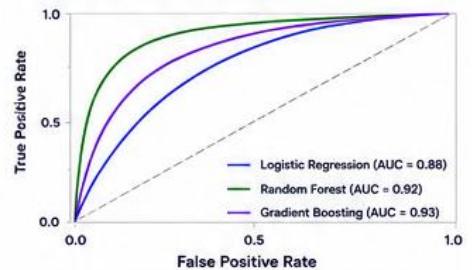
BEST MODEL: GRADIENT BOOSTING CLASSIFIER

Achieved the highest accuracy and ROC-AUC score, performing best across all metrics with strong generalization.

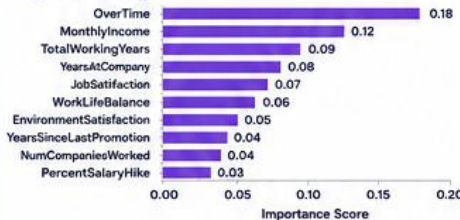
6. CONFUSION MATRIX (Gradient Boosting)

		Actual	
		No	Yes
Predicted	No	236	11
	Yes	19	28

7. ROC CURVE COMPARISON



8. FEATURE IMPORTANCE (TOP 10) (Gradient Boosting)



9. ATTRITION INSIGHTS

- Overtime is the most significant factor affecting attrition.
- Lower income and fewer years at the company increase risk.
- Low job satisfaction and poor work-life balance drive employees to leave.
- Retention strategies should focus on early-career employees.

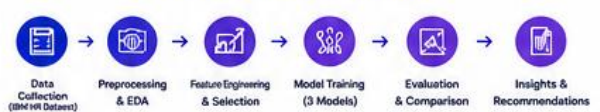
10. BUSINESS RECOMMENDATIONS

- Improve work-life balance and reduce overtime stress.
- Focus on employee engagement and job satisfaction programs.
- Provide competitive compensation and career growth.
- Monitor at-risk employees using predictive analytics dashboards.
- Strengthen retention strategies for critical roles and early talent.

11. PROJECT DELIVERABLES

- analysis.ipynb – Jupyter Notebook
- HR_Attrition.csv – Dataset Used
- summary.pdf – Project Summary
- charts/ – Visualizations (PNG)

12. MACHINE LEARNING PIPELINE



PROJECT ARCHITECTURE



TECH STACK



CONCLUSION

This project demonstrates the end-to-end process of building an employee attrition prediction model using machine learning. The Gradient Boosting Classifier achieved the highest performance with ROC-AUC of 0.93. The insights generated can help HR teams reduce attrition and improve employee retention.

CONTACT

9493929098
pokalaravindranathtagoor@gmail.com
Hyderabad, Telangana, India
LinkedIn: pokalaravindranathtagoor
GitHub: POKALARAVINDRANATHTAGOOR



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DATA. INSIGHT. IMPACT.